# RAHUL TANGSALI

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## Education

**University of Southern California - M.S. in Computer Science (08/2023 - 05/2025)** Los Angeles, CA
**Pune Institute of Computer Technology - B.E. in Computer Engineering (08/2019 - 06/2023)** Pune, India

## Technical Skills

**Languages:** Python, TypeScript, JavaScript, C++, SQL, Swift
**Frameworks:** React, Next.js, Angular, React Native, Node.js, Express.js, FastAPI
**Databases:** PostgreSQL, MySQL, MongoDB, Pinecone, Supabase
**AI/ML:** LangGraph, LangChain, RAG, LLMs, MCP, NLP, Deep Learning
**Cloud/Tools:** AWS, GCP, Git, GitHub, Docker, Vercel, Railway

## Work Experience

**OneHermes** Madison, WI
Founding Software Engineer 06/2026 – Present

- Shipped **300+ tickets** regarding new features, production enhancements and defect fixes for an enterprise workforce-management platform using **Angular, Node.js, Express, and MongoDB**, across scheduling, leave, timesheets, and workforce workflows.
- Engineered scheduling logic for **shift conflicts, overnight clock-in/out, role-based approvals, and real-time status synchronization** across employee, manager, and HR workflows.
- Developed and tested **EVV integrations and Patient Care workflows**, implementing API-driven healthcare data flows, validation, and frontend/backend functionality for client requirements.
- Worked directly with stakeholders across **3+ customer environments**, owning reported issues from **requirements and root-cause analysis through implementation, testing, release validation, and client follow-up**.

**University of Southern California** Los Angeles, CA
Research Software Engineer 08/2025 – 06/2026

- Built **Tommy**, an AI-powered virtual assistant deployed on the USC OIS website to answer F-1/J-1 immigration questions **24/7**, handling **2,000+ student queries/month** and reducing routine inquiry load on advising staff.
- Developed automated data-ingestion and web-scraping pipelines across **15+ university and government sources** to maintain an up-to-date knowledge base for immigration-related responses.
- Designed a **human-in-the-loop feedback pipeline** using thumbs up/down signals and logged conversations to identify response failures and iteratively improve chatbot quality across model and knowledge-base updates.

**Sumeru Inc.** Los Angeles, CA
Full Stack Developer Intern 01/2025 – 05/2025

- Migrated a client platform serving **25,000+ monthly active users** from legacy PHP to a modern **Next.js and Node.js** architecture, implementing reusable frontend components, backend APIs, and business logic.
- Reduced **P95 API response time from 1.8s to 650ms** under high traffic by optimizing backend request handling, database queries, and schema design.
- Worked directly with client stakeholders to translate product requirements and feedback into user stories, application flows, and production features, iteratively refining functionality based on testing and customer feedback.

**Sara Technology Inc.** Los Angeles, CA
Software Engineer Intern 05/2024 – 08/2024

- Built onboarding and interactive speech-practice workflows using **React Native**, animations, and audio for a speech-therapy application deployed to **3 schools and 5 therapy clinics within 2 months**.
- Developed interactive speech-therapy games with **real-time audio playback, progress tracking, and animated transitions** to support engaging mobile learning experiences.
- Contributed to the company's **Next.js and TypeScript** website and product experience; subsequent company milestones included **$80K in investor funding** and selection for a **Columbia Business School grant**.

## Projects

**MedAssist AI – Full-Stack Medical Assistant** *LangGraph, FastAPI, Next.js, Pinecone, OpenAI*

- Built an **agentic healthcare assistant** using **LangGraph and FastAPI** with **8 dynamically routed tools** integrating NIH, FDA, and NPI APIs, achieving **94.3% tool-routing accuracy** across 35 synthetic evaluation cases.
- Engineered a cascading retrieval architecture combining **specialized APIs, Pinecone RAG over 23,489 document chunks, and LLM fallback**, with source citations surfaced for retrieved information.
- Built an **LLM-as-Judge evaluation framework** across 11 categories, achieving **4.37/5 correctness**; shipped a **Next.js** frontend with JWT authentication, user profiles, and real-time SSE response streaming.

**RAG-Powered Document Research Assistant** *Next.js, LangGraph, LangChain, OpenAI, Pinecone, Cohere*

- Built a **full-stack RAG application** using **Next.js and LangGraph** for conversational document querying, achieving **93.75% faithfulness** and **97.32% answer correctness** across a 20-case LLM-as-Judge evaluation suite.
- Engineered a multi-stage retrieval pipeline using **OpenAI embeddings, Pinecone vector search, Cohere reranking, LLM-based document grading, and web-search fallback**, with SSE streaming and MCP tool integrations.